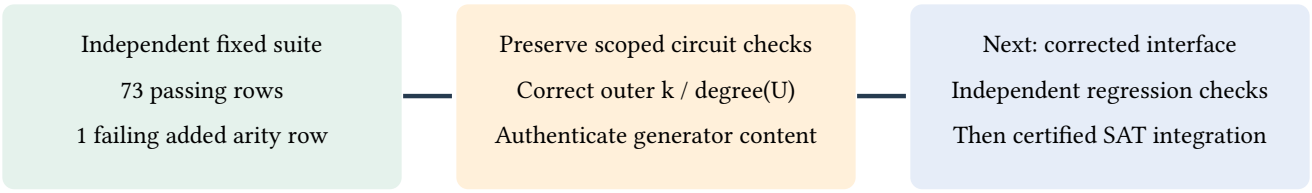
Rationality software: independent review

DEC-20261031-cd6460 · TASK-20261031-25b2d5



TASK-20261031-60575a · one invocation · assertion counts, not performance units

The independent review confirms the prescribed symbolic rationality circuits and signed outputs, and identifies one missing outer-arity check. Decision: refine the standalone interface before SAT integration. Preserve the reviewed builders and all original evidence.

One independent fixed suite ran once: 73 rows passed and one additional deterministic row failed. Every prescribed fixture and mandatory mutation passed. The thirteen scientific input hashes were unchanged. The symbolic nilpotent counts for k=0 through5 are0,3,9,15,21,36 auxiliaries and equations; guard uses10 auxiliaries/12 equations, slack9/8. These are field-generator counts, not solving work.

R1 is an explicit counterexample. Over F5 on $y^2=x^3+1$, take target(0,1), interval t=0,s=1, A=−X, B=X and $U=X^4$. Starting with the valid guarded finite-four chart, change both copies of the outer k tag from4 to3. The candidate checker accepts and returns four copies of(0,1) plus identity padding. It compares metadata copies but does not independently enforce k=degree(U) on this branch. A separately prevalidated source chart excludes this mutation; the standalone checker does not. Require an exact integer k in0..5 and equality with the monic quotient degree before stratum selection.

Five review joints hold at their stated scope: symbolic circuits, rationality predicates, signed fidelity, execution completeness and claim boundaries. Strict assignment validation breaks on R1. Both F1 guard roots preserve the same divisor4(0,1)+(0,4). The review retains repeated roots, branch roots, zero-B charts, wrapped intervals and targets outside the interval.

The generic IR evaluator checks supplied equations; it does not authenticate their meaning. In the reviewer's specialization control, the generic evaluator accepts the altered program while the independent exact-symbolic comparator rejects it. The reviewed builder output passes that comparator. SAT integration must regenerate or otherwise authenticate the exact canonical generators from trusted public parameters before accepting evaluated witnesses. Retain a separate evaluator and charge authentication as well as residual and certificate checking.

Execution custody is explicit. The first checkpoint operation reached a60-second wrapper watchdog before any suite launched. After root and reviewer confirmed no surviving children, one authorized retry used an1800-second gate-only watchdog and returned zero. Scientific assertions and the240-second/2GiB suite limits were unchanged. Both checkpoint attempts, original wrapper, exact command, stdout/stderr and the failed assertion remain in the raw record. No unchanged rerun was performed. The native gate passed declared review-plan/attestation consistency; it does not prove software correctness or authenticate serving model and effort.

Next admit an additive corrected interface, retaining the frozen failing source and test. Verify arity and canonical-equation rejection independently, then connect to certified SAT with complete chart handling. Compare identical complete projected model sets and charge witness reconstruction and certificate checking. This decision adds no solver run, census, benchmark, speed claim or regularity claim.

Evidence publication: 0f1432c952bd46add0557581d98eb0f697e1f59d. Exact sources: report-inputs.json.